Referee report

"The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane $\operatorname{Re} s > 7/8$"
(openai/math, commit `adc7f1241b42`, preprint dated 30 September 2026)

Independent verification report

7 October 2026

1 Summary of the submission

The preprint claims that every Dirichlet L-function, including $\zeta(s)$, and every finite-order Hecke L-function over $F = \mathbb{Q}(\sqrt{-3})$ has no zero in $\operatorname{Re} s > 7/8$ (principal pole at $s = 1$ excepted). The argument is a family-wide bootstrap. Let β_* be the supremum of the real parts of the zeros of all primitive finite-order Hecke L-functions over F. If $\beta_* > \sigma_0$, a probe $J_\eta(Z)$ built from cubic-theta (Kubota–Patterson) coefficients averaged against sextic residue characters is bounded directly (the *low* estimate, via cubic reflection and large sieves). The same probe is compared, after Poisson summation, with the Mellin signal of $1/L(s,\eta)$ (the *high* estimate). The non-principal Poisson rows are Hecke twists inside the same family, whose zeros satisfy $\operatorname{Re}\rho \le \beta_*$; rows carrying zeros are counted by a zero detector together with sextic large-sieve and moment estimates. A Landau-type continuation principle (Proposition 2.1) then contradicts the definition of β_*. Part I gives $11/12$, and Part II bootstraps from it to $7/8$. A Lean 4 formalization of the ζ and Dirichlet statements accompanies the preprint.

2 Overall assessment

Verdict. The main theorem is correct for ζ and for all Dirichlet L-functions. I rebuilt the accompanying Lean 4 formalization locally: 2,924 modules, no errors. The resulting theorem `OAI.riemannZeta_ne_zero_of_seven_eighths_lt_re`, whose type matches the comparator challenge verbatim, depends only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The remaining trust base is the Lean kernel, the official Mathlib build cache, and Mathlib's definitions of `riemannZeta` and `DirichletCharacter.LFunction`.

I found no structural defect that would refute the argument on general grounds, and no error in any part I checked in full. The portions I verified by hand or by exact computer algebra are the continuation criterion, the Hecke-to-Dirichlet transfer, the exponent bookkeeping of Section 20, and the endpoint certificate (Lemma 20.2). The technically heaviest parts I did *not* verify by hand: Proposition 5.1, Lemma 9.1, Propositions 9.2 and 19.2, and Sections 17–18, about 10^4 lines of TEX. For these, the evidence of correctness is the formal verification reported in Section 4, which certifies the theorem but not the prose argument line by line.

3 What was checked

1. **Provenance.** The repository, PDF (199 pp.), TEX source (16,677 lines) and Lean sources were fetched through the GitHub API, and SHA-256 hashes were recorded.

2. **Structure.** 67 numbered statements were extracted from the TEX labels and references. The section-level citation graph is acyclic; all 11 forward references are announcements of later proofs.

3. **Barrier tests.**

   - *Euler product.* The argument uses Euler products essentially: through the local Euler identity (Lemma 7.x, `lem:local-euler`), through Möbius coefficients in the inverse detector polynomials, and through closure of the family under twisting. It therefore does not apply to the Davenport–Heilbronn function. We re-computed that function's off-line zero $0.808517 + 85.699348i$.
   - *Single zero versus density.* The target zero is never counted. It enters through the size of the Mellin signal of $1/L$, and density-type counts are used only for the other rows. This is a legitimate use of density estimates.
   - *Uniformity.* All exponent margins are fixed before the target: $m_{\mathrm{lo}} = \Delta$, $m_{\mathrm{hi}} = \frac{51}{64}\Delta$, $m_{\mathrm{small}} = 63/800$, $m_w = 23/960$, $m_z = 13/9600$, and κ_P.
   - *Consistency.* The consequences (no Landau–Siegel zeros in $(7/8, 1)$, Vinogradov's least non-residue conjecture via Bhargava–Ivanyos–Mittal–Saxena) contradict no known theorem or numerical data.

4. **Exact arithmetic.** 66 explicit identities and inequalities were verified with `sympy` and `mpmath`; all pass. They include the boundary identity $C_{\mathrm{II}}(s) = l_x/2 + s - 1 + h/6 = s - 11/16$, the completed-square certificate of Lemma 20.2, and every rational constant in Section 20. A 401×401 grid gives $\min(-E_*) = 2.28 \times 10^{-4}$, about twice the certified $49/440640$.

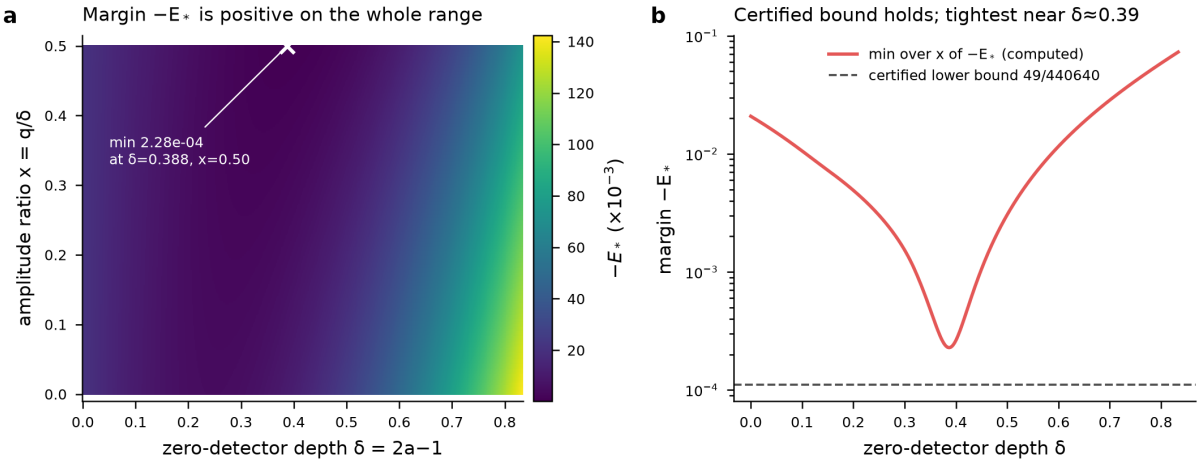


Figure 1: Lemma 20.2. (a) The margin $-E_*(\delta, x)$ over the full parameter range. (b) Its minimum over x against the certified lower bound $49/440640$ (dashed). The certificate holds everywhere and is tightest near $\delta \approx 0.39$, $x = 1/2$.

4 Formal verification

- **Statement fidelity.** The challenge file states `theorem riemannZeta_ne_zero_of_seven_eighths_lt_re {s : ℂ} (hs : (7 / 8 : ℝ) < s.re) : riemannZeta s ≠ 0` using Mathlib definitions only. A verbatim copy of this statement was proved from the solution theorem, which confirms the type match; the same holds for the Dirichlet statement. Mathlib's junk value `riemannZeta 1` $= (\gamma - \log 4\pi)/2 \neq 0$ is consistent with the statement.

- **Build.** Lean 4.34.1 (official `darwin_aarch64` release) with Mathlib `d13f23b` (official cache), PrimeNumberTheoremAnd `c39a751` and rellich-kondrachov `70f85d4` (with the repository's compatibility patches). The command `lake build OAI.NumberTheory.DirichletL.Nonvanishing` completed 7,061 jobs, including local elaboration and kernel checking of all 2,924 OAI modules, with no errors and no `sorry` warnings (252 min on 8 cores / 16 GB).
- **Axioms.** `#print axioms` gives `[propext, Classical.choice, Quot.sound]` for both the ζ and the Dirichlet theorem.
- **Static scan.** The OAI closure contains no occurrence of `sorry`, `axiom`, `native_decide`, `implemented_by`, `extern`, `unsafe`, `opaque`, `run_cmd`, `elab`, `macro_rules`, `initialize`, `skipKernelTC` or `addDecl`. PrimeNumberTheoremAnd contains two test axioms in a tactic file; the axiom output shows that the theorem does not use them.
- **Independent kernel replay.** Time-boxed. A full `leanchecker` replay of all 2,924 OAI modules was stopped after about 60 minutes without completing; no error had been reported. The replay of the two final modules (`Nonvanishing` and `FinalAssemblyUnconditional`) completed without error. In the ordinary build, the kernel already checks every declaration as it is added.
- **Not covered.** The Hecke challenge, which uses local definitions, was not built, and no mapping from the paper's numbered statements to Lean declarations is provided.

5 Major comments

1. **Where the boundary comes from.** The low estimate and the Mellin normalization give exactly $\sigma_0 = 1 - h/6 + b/12$, which yields $11/12$ for $(h, b) = (1/2, 0)$ and $7/8$ for $(13/16, 1/8)$. Several stated hypotheses stop exactly at $7/8$:
   - `lem:bin-contour-accounting` assumes $\sigma_0 \in [7/8, 1)$;
   - the second region of `lem:local-euler` requires $\operatorname{Re} x \geq 7/8$;
   - `lem:plain` assumes $\kappa \geq 3/4$.

   The authors should say explicitly that $7/8$ is (nearly) optimal for this parametrization. Our re-optimization of the Section 20 constraints gives 0.87496; see Section 7.
2. **Hecke statement.** The Lean challenge for Hecke L-functions over $\mathbb{Q}(\sqrt{-3})$ uses definitions local to the challenge file. Their faithfulness to the paper's $L_F(s, \eta)$ should be audited independently. The ζ and Dirichlet challenges use only Mathlib definitions.
3. **Dependency patches.** The build applies large compatibility patches to third-party packages (e.g. 2 MB for PrimeNumberTheoremAnd). Soundness is protected by the axiom check, but the authors should state what the patches change.

6 Minor comments

1. Line 16141: the bound $R_* + \Delta/4 \leq 139/96$ is valid, but the exact maximum is $17/12 + 1/96 = 137/96$.
2. Corollary 1.2: please quote the precise hypothesis of [1, Conj. 6.3, Thm. 6.7] being verified (we did not check the cited source).

7 On extending the method to the Riemann hypothesis

The method cannot reach $\operatorname{Re} s > 1/2$. Under the paper's normalization $M + \ell = 1$ (satisfied in both parts):

- *Low-side geometry.* $\sigma_0 = \frac{2}{3} + \frac{l_x}{2} + \frac{b}{4} \geq \frac{2}{3}$ for every geometry. The $1/6$ in $1 - h/6$ comes from the pole of $\zeta_F(6z)$, that is, from the sextic frequency structure ua^6.
- *Density barrier.* Rows whose zeros all lie near $\operatorname{Re} s = 1/2$ admit no counting saving ($R = 1$). Inserting $R = 1$ into the high exponent forces $\frac{5}{2} l_x + \frac{13}{12} b > 1$, hence $\sigma_0 > 13/15 \approx 0.8667$. This holds even with density-hypothesis-strength counts $R = 1 - \delta$.
- *Bootstrap saturation.* Re-running Part II with the improved a priori bound $\alpha = 3/4$ gives at most 0.87451. The iteration has a fixed point at about 0.8745.

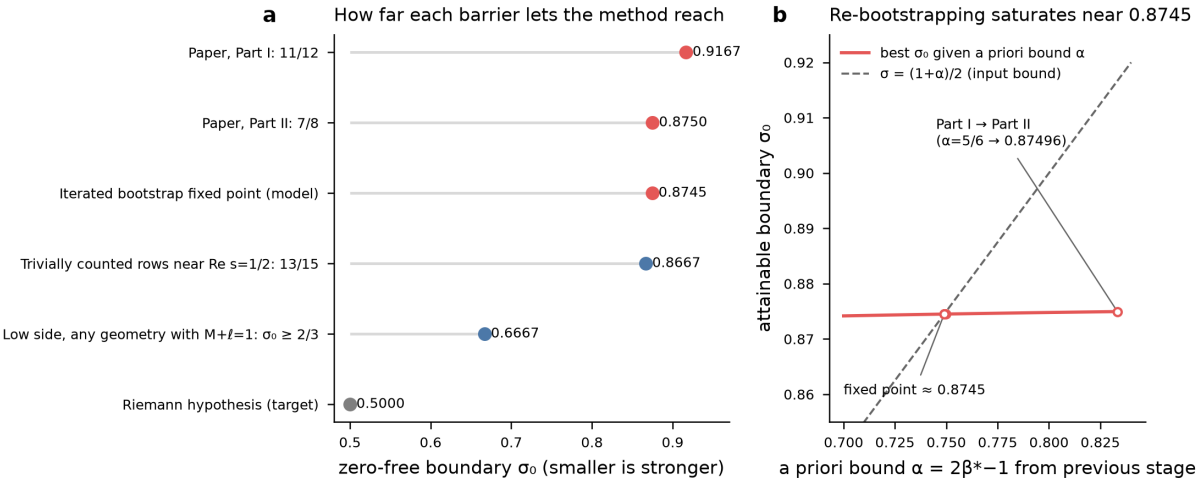


Figure 2: (a) The boundaries reachable under each barrier. (b) The best boundary as a function of the a priori bound $\alpha = 2\beta_* - 1$ from the previous stage; iterating the bootstrap saturates near 0.8745. Both panels use the paper's exponent formulas extended verbatim beyond their stated ranges, so they give optimistic bounds.

8 Requests to the authors

1. Provide a mapping from the paper's numbered statements to Lean declaration names, at least for Proposition 5.1, Lemma 9.1, Lemma 17.1, Lemma 18.1 and Proposition 19.2.
2. Publish a comparator log (including `nanoda` or `lean4checker` replay) for the three challenges.
3. Clarify the role of the dependency patches and of the Hecke challenge definitions.

Limitations of this report

This report does not replace expert peer review. A planned independent LLM-based second opinion on each lemma could not be run because of a usage limit. The barrier and improvement analyses are model computations built on the paper's explicit exponent formulas; they do not establish that the corresponding lemmas extend.

References

[1] A. Bhargava, G. Ivanyos, R. Mittal, N. Saxena, as cited in the preprint (not checked).